$\tilde{\theta} = h(\bar{X}) = h\left(\frac{1}{n}\sum_{i=1} X_i\right)$

Т5

$U_{\theta, 2\theta}$ - равном. распр на $[\theta, \theta \cdot 2]$

а) $EX_1 = \frac{3\theta}{2}$ $\theta = \frac{2}{3} EX_1$

$\tilde{\theta} = \frac{2}{3}\bar{X}$

найдем оценку метода моментов $\tilde{\theta}_k$ по k-му моменту

$EX_1^k = \int_0^\theta y^k \frac{1}{\theta} dy = \frac{\theta^k}{k+1}$

$\theta = \sqrt[k]{(k+1)EX_1^k}$

$\tilde{\theta}_k = \sqrt[k]{(k+1)\bar{X}^k}$

■ Оценкой максимального правдоподобия (ОМП) $\hat{\theta}$ для неизвестного θ называют такое θ, при котором достигается максимум ф-ии $f(\bar{X}, \theta)$

$p(x) = \frac{1}{2\theta - \theta} = \frac{1}{\theta}$

$L(\theta) = \left(\frac{1}{\theta}\right)^n = \frac{1}{\theta^n}$

$\theta \le X_i \le 2\theta$

- $\theta \le X_i$ $(X_{(1)})$ - мин. эл-т. выборки
- $X_i \le 2\theta \Rightarrow \theta \ge \frac{X_i}{2}$ $(X_{(n)})$ - макс. эл выборки

$\frac{X_{(n)}}{2} \le \theta \le X_{(1)}$, $L(\theta) = \frac{1}{\theta^n}$ макс при мин. θ

$\hat{\theta} = \frac{X_{(n)}}{2}$

б) $E\hat{\theta} = \frac{1}{2} E[X_{(n)}] = \frac{1}{2}\int_{\theta}^{2\theta} t \cdot \frac{n}{\theta}\left(\frac{t}{\theta}-1\right)^{n-1} dt =$

$= \left\{ \begin{matrix} u = \frac{t}{\theta} \\ du = \frac{1}{\theta} dt \end{matrix} \right\} = \frac{\theta}{2}\int_1^2 n u (u-1)^{n-1} du$

$\int_1^2 u \cdot n(u-1)^{n-1} du = \int_1^2 u\, d((u-1)^n) = u(u-1)^n \Big|_1^2$

$- \int_1^2 (u-1)^n du = \left(u(u-1)^n - \frac{(u-1)}{n+1}\right)\Big|_1^2 =$

$= 2 - \frac{1}{n+1}$

$E\tilde{\theta} = \theta - \frac{\theta}{2(n+1)} = \frac{\theta(2n+1)}{2n+2} \Rightarrow \tilde{\theta}$ - смещенная

$\tilde{\theta}' = \frac{2n+2}{2n+1}\tilde{\theta}$ - исправленная

$D\tilde{\theta} = E\tilde{\theta}^2 - E^2\tilde{\theta}$

$E\tilde{\theta}^2 = \frac{1}{4}\int_{\theta}^{2\theta} t^2 \cdot \frac{n}{\theta}\left(\frac{t}{\theta}-1\right)^{n-1} dt = \left\{ v = \frac{t}{\theta} \right\}$

$$= \frac{\theta^2}{4}\int_1^2 n v^2 (v-1)^{n-1} dv = 4 - \frac{2}{n+1}\left(2 - \frac{1}{n+2}\right) =$$

$$= \frac{4n^2+8n+2}{n^2+3n+2}$$

$$E\tilde{\theta}^2 = \frac{\theta^2}{4}\cdot\frac{4n^2+8n+2}{n^2+3n+2}$$

$$D\tilde{\theta} = \theta^2\left(\frac{2n^2+4n+1}{(2n+4)(n+1)} - \left(\frac{2n+1}{2n+2}\right)^2\right) =$$

$$= \frac{\theta^2 n}{4(n+1)^2(n+2)}$$

$$D\tilde{\theta} = \frac{4(n+1)^2}{(2n+1)^2}\cdot\theta^2\cdot\frac{n}{4(n+1)^2(n+2)} \xrightarrow[n\to\infty]{} 0$$

$\Rightarrow \tilde{\theta}$ – состоят.

$$E\tilde{\theta} = \frac{2}{3}E\bar{X} = \frac{2}{3}E\xi = \frac{2}{3}\cdot\frac{3}{2}\theta = \theta \Rightarrow$$

$\tilde{\theta}$ – несмещ.

$$D\tilde{\theta} = \frac{4}{9}D\bar{X} = \frac{4}{9n}D\xi$$

$$\int_\theta^{2\theta}\frac{x^2}{\theta}dx = \frac{1}{\theta}\cdot\frac{x^3}{3}\Big|_\theta^{2\theta} = \frac{1}{\theta}\left(\frac{8\theta^3}{3} - \frac{\theta^3}{3}\right) = \frac{7}{3}\theta^2$$

$$D\tilde{\theta} = \frac{4}{9n}\cdot\left(\frac{7}{3}\theta^2 - \frac{9}{4}\theta^2\right) = \frac{\theta^2}{12}\cdot\frac{4}{9n} = \frac{\theta^2}{27n} \xrightarrow{n\to\infty} 0$$

$\tilde{\theta}$ – самостоятельна

c) $\frac{\theta^2}{27n} \geq \frac{n\theta^2}{(n+2)(2n+1)^2} \Rightarrow$

оценка методом правдоподобия эффектив-
ная

d) X_i имеют распределение $U_{\theta, 2\theta}$

$Y_i = \frac{X_i}{\theta} - 1$ имеют распр. $U_{0,1} \Rightarrow$

$Y_{(n)} = \max\{Y_1, \dots, Y_n\} = \frac{\max\{X_1, \dots, X_n\}}{\theta} - 1 =$

$= \frac{X_{(n)}}{\theta} - 1 = G(\bar{X}, \theta)$

величина $Y_{(n)}$ имеет не зависящую от
θ ф-ию распр.

$$F_{Y_{(n)}}(y) = P(\eta < y) = \begin{cases} 0, & y < 0 \\ y^n, & y \in [0,1] \\ 1, & y > 1 \end{cases}$$

$P(g_1 < G(\bar{X}, \theta) < g_2) =$

$= P\left(\frac{X_{(n)}}{g_2 + 1} < \theta < \frac{X_{(n)}}{g_1 + 1}\right)$

$1 - \varepsilon = P(g_1 < Y_{(n)} < 1) =$

$= F_{Y_{(n)}}(1) - F_{Y_{(n)}}(g_1) = 1 - g_1^n$

$g_1 = \sqrt[n]{\varepsilon}$ получим:

$$1-\varepsilon = P(\sqrt[n]{\varepsilon} < Y_{(n)} < 1) = P\left(\frac{X_{(n)}}{2} < \theta < \frac{X_{(n)}}{\sqrt[n]{\varepsilon}}\right)$$

е) $EX_1 = \frac{3}{2}\theta$

$DX_1 = \frac{\theta^2}{12}$

$$G(\bar{X}, \theta) = \sqrt{n}\,\frac{\bar{X} - EX_1}{\sqrt{DX_1}} = \sqrt{n}\,\frac{\bar{X} - 1{,}5\theta}{\sqrt{\frac{\theta^2}{12}}}$$

$$G(\bar{X}, \theta) = \sqrt{12n}\,\frac{\bar{X} - 1{,}5\theta}{\theta} = \sqrt{12}\,n\left(\frac{\bar{X}}{\theta} - 1{,}5\right)$$

слабо сход. к $N(0,1)$

$\eta \sim N(0,1)$

$\exists\, C = \tau_{1-\frac{\varepsilon}{2}}$

$$-C < \sqrt{12n}\left(\frac{\bar{X}}{\theta} - \frac{3}{2}\right) < C$$

$$-\frac{C}{\sqrt{12n}} < \frac{\bar{X}}{\theta} - \frac{3}{2} < \frac{C}{\sqrt{12n}}$$

$$1{,}5 - \frac{C}{\sqrt{12n}} < \frac{\bar{X}}{\theta} < \frac{3}{2} + \frac{C}{\sqrt{12n}}$$

$$\frac{\bar{X}}{1{,}5 + \frac{C}{\sqrt{12n}}} < \theta < \frac{\bar{X}}{1{,}5 - \frac{C}{\sqrt{12n}}}$$

$$\left(\frac{\bar{X}}{1{,}5 + \frac{C}{\sqrt{12n}}};\ \frac{\bar{X}}{1{,}5 - \frac{C}{\sqrt{12n}}}\right)$$